SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5

Duration :60 Minutes

Code of Conduct
I YDDALA KUSHAL KUMAR(192524068) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: YADDALA KUSHAL KUMAR

Assessment Tasks
Industry Problem Solving Task

1. Retail Data Optimization Problem

Objective: Build a big data architecture for real-time inventory tracking, demand forecasting, and personalized marketing.

Architecture:

```
POS + E-Commerce + Mobile App + IoT Sensors
          ↓
     Apache Kafka
          ↓
  ┌───────────────┴───────────────┐
  ↓            ↓
Real-Time Processing    Data Lake
Apache Spark/Flink      HDFS/S3
  ↓            ↓
Inventory Dashboard     Historical Data
  ↓            ↓
  └───────────────┬───────────────┘
          ↓
    ML / Analytics Layer
          ↓
Demand Forecasting + Recommendation Engine
          ↓
   Managers + Customers + Marketing
```

Tools and technologies:

- **Apache Kafka:** Collects sales, inventory, website, and customer events in real time.
- **HDFS / Amazon S3:** Stores large volumes of structured and unstructured historical data.
- **Apache Spark:** Performs large-scale batch and real-time analytics.
- **Apache Flink:** Can be used for low-latency stream processing.
- **NoSQL databases:** MongoDB/Cassandra can store rapidly changing inventory and customer information.
- **Power BI/Tableau:** Provides dashboards for inventory and sales monitoring.

Analytics models:

- **ARIMA/Prophet/LSTM:** Demand forecasting based on historical sales, seasonality, and trends.
- **Collaborative Filtering:** Recommends products based on similar customer behavior.
- **Clustering:** Groups customers according to purchasing behavior.
- **Association Rule Mining:** Identifies products frequently purchased together.

Result: The architecture provides real-time stock visibility, predicts future demand, reduces overstocking and stockouts, and provides personalized recommendations.

2. Social Media Fraud Detection Problem

Objective: Detect fake accounts, fraudulent activities, and misinformation with minimum delay.

Architecture:

Posts + Comments + Likes + Shares + User Profiles
↓
Apache Kafka
↓
Stream Processing
Spark Streaming/Flink
↓
Feature Extraction
↓
ML Detection Models
↓ ↓
Fake Account Misinformation
↓ ↓
Alert System
↓
Moderator/Dashboard

Technologies:

- **Apache Kafka:** Handles billions of user events and provides scalable event ingestion.
- **Spark Streaming / Apache Flink:** Processes interactions continuously.
- **Hadoop/S3:** Stores historical social media data.
- **NoSQL databases:** Store user profiles, interaction graphs, and real-time features.
- **Elasticsearch:** Enables fast searching and investigation of suspicious activities.

Machine Learning:

- **Random Forest/XGBoost:** Classifies users or posts as legitimate or suspicious.
- **Logistic Regression:** Provides a simple baseline classification model.
- **Graph Neural Networks (GNNs):** Analyze relationships between users and identify coordinated fake-account networks.
- **NLP models:** Analyze text to identify potentially misleading or harmful information.
- **Anomaly Detection:** Identifies unusual activity such as thousands of likes or shares in a short period.

Role of stream processing: Stream processing analyzes events as they arrive rather than waiting for daily batch processing. This enables rapid detection and alerts.

Result: The platform can identify suspicious accounts and misinformation in near real time while scaling to billions of interactions.

3. Government Data Platform Problem

Objective: Integrate census, economic, and geospatial data while maintaining interoperability, privacy, and security.
Architecture:

Census Data
Economic Data → Data Ingestion → Data Lake
Geospatial Data
↓
Data Integration
↓
Government Data Warehouse
↓
Analytics / GIS / Reports
↓
Authorized Departments

Data management strategy:

1. **Data Lake:** Store raw census, economic, geospatial, and departmental datasets.
2. **Data Warehouse:** Store cleaned and integrated datasets for reporting and analysis.
3. **ETL/ELT:** Use tools such as Apache NiFi, Spark, or cloud data integration services to extract, transform, and load data.
4. **Common data standards:** Define common formats, schemas, metadata, identifiers, and APIs so different departments can exchange information.
5. **Master Data Management:** Maintain consistent records for common entities such as citizens, locations, organizations, and government services.
6. **Metadata Management:** Maintain information about data ownership, source, quality, meaning, and usage.
7. **Data Quality:** Apply validation, deduplication, consistency checks, and missing-value handling.

Security and privacy:

- Role-Based Access Control (RBAC)
- Encryption at rest and in transit
- Multi-factor authentication
- Data masking and anonymization
- Audit logs
- Data classification
- Consent and purpose-based access where applicable

Governance: A central data governance committee should define policies for data ownership, access, quality, retention, privacy, and compliance.
Result: Departments can securely share standardized and integrated information while protecting sensitive citizen data.

4. Banking Fraud Detection Problem
Objective: Detect fraudulent transactions in real time with high accuracy and very low latency.
Architecture:

ATM + POS + Mobile Banking + Internet Banking
↓
Apache Kafka
↓
Real-Time Stream Processing
Apache Flink / Spark
↓
Feature Engineering
↓

↓ ML Fraud Model ↓ Rule Engine

XGBoost/Random Forest   Transaction Rules

↓   ↓

↓

Fraud Risk Score

↓

Approve / Reject / Alert

↓

Historical Data Lake

Technologies:

- **Kafka:** Ingests millions of transactions continuously.
- **Apache Flink:** Performs low-latency transaction stream processing.
- **Spark:** Performs historical analysis and model training.
- **Data Lake:** Stores historical transactions for analysis and training.
- **NoSQL/Redis:** Provides fast access to customer profiles and recent transaction features.

Algorithms:

- **XGBoost:** High-performance classification of fraudulent transactions.
- **Random Forest:** Robust fraud classification model.
- **Logistic Regression:** Useful as a baseline model.
- **Isolation Forest:** Detects unusual transaction patterns.
- **Autoencoders:** Detect complex anomalies using neural networks.
- **Graph analytics:** Detects relationships between suspicious accounts, devices, cards, and merchants.

Important features:

- Transaction amount
- Transaction location
- Time of transaction
- Frequency of transactions
- Device information
- Merchant category
- Previous transaction behavior
- Sudden changes in spending patterns

Result: Each transaction receives a fraud-risk score within milliseconds. High-risk transactions can be blocked or sent for additional verification.

5. Agriculture Smart Farming Problem

Objective: Use distributed sensor, weather, drone, and crop data to improve agricultural productivity and resource utilization.

Architecture:

Soil Sensors + Weather Stations + Drones + Satellites

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IoT Gateway

↓

MQTT / Apache Kafka

↓

Stream Processing
Spark / Apache Flink

↓

Data Lake

↓

Analytics + ML Models

↓

↓   ↓   ↓

Crop Prediction   Irrigation Optimization   Disease Detection

↓ ↓ ↓

↓

Farmer Dashboard

Technologies:

- **IoT sensors:** Collect soil moisture, temperature, humidity, pH, and other field measurements.
- **MQTT:** Efficiently transfers sensor data.
- **Kafka:** Handles large-scale real-time agricultural data.
- **HDFS/S3:** Stores historical sensor, drone, weather, and satellite data.
- **Spark/Flink:** Processes large-scale and streaming agricultural data.
- **GIS:** Maps field conditions and identifies areas requiring attention.

Analytics models:

- **Regression:** Predict crop yield using soil, weather, and historical data.
- **Random Forest/XGBoost:** Predict crop health and disease risk.
- **CNN:** Analyze drone or satellite images for crop disease and vegetation health.
- **Time-series models:** Predict weather, soil moisture, and irrigation requirements.
- **Clustering:** Divide fields into zones with similar soil and crop characteristics.

Decision-making applications:

- **Smart irrigation:** Water is supplied according to soil moisture and crop requirements.
- **Fertilizer optimization:** Fertilizer is applied only where required.
- **Disease detection:** Drone images can identify unhealthy crop regions.
- **Yield prediction:** Farmers can estimate expected production before harvesting.
- **Weather-based decisions:** Farmers can adjust irrigation and harvesting schedules according to weather forecas